# EXTENDING A RATIONAL MAP BY NORMALIZED BLOWUPS FOR A NAGATA SURFACE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the proper modification over which the rational map is defined, and its domination by a normalization followed by normalized blowups in closed points.

*Editorial note.* The label lemma-extend-rational-map-normalized-blowing-up names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Editorial note.* Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-extend-rational-map-normalized-blowing-up –start 1389 –end 1411 –local-id 28122 returned RESULT=OK (L28122 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

*Editorial note.* The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

*Allowed standard prerequisites (editorial).*

- Extending a rational map after a modification. Divisors, the lemma producing a proper modification over which a rational map into a proper target is defined. Produces the proper morphism to the source which is an isomorphism over the domain of the rational map and carries a morphism agreeing with it.
- Dominating a proper modification by normalized blowups. Same chapter, the lemma dominating a proper modification of a Nagata surface by a sequence of normalized blowups in closed points. Turns the modification into a normalization followed by a sequence of normalized blowups and provides the morphism to the target.

*Source reference.* Original lemma 0C4V: sources/stacks-chow/tags.txt, source lines 15240–15240. Prerequisite pointer invoked at proof line(s) 1405; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0BBT: sources/stacks-resolve/pinned-original-source.tex, source lines 1277–1277. Prerequisite pointer invoked at proof line(s) 1408; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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*Date*: stacks-project:resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X$ be a scheme over $S$ which is Noetherian, Nagata, and has dimension 2. Let $Y$ be a proper scheme over $S$. Given an $S$-rational map $f : U \to Y$ from $X$ to $Y$ there exists a sequence*

$$X_n \to X_{n-1} \to \ldots \to X_1 \to X_0 \to X$$

*and an $S$-morphism $f_n : X_n \to Y$ such that $X_0 \to X$ is the normalization, $X_{i+1} \to X_i$ is the normalized blowing up of $X_i$ at a closed point, and $f_n$ and $f$ agree.*

*Proof.* Applying Divisors, Lemma 0C4V we find a proper morphism $p : X' \to X$ which is an isomorphism over $U$ and a morphism $f' : X' \to Y$ agreeing with $f$ over $U$. Apply Lemma 0BBT to the morphism $p : X' \to X$. The composition $X_n \to X' \to Y$ is the desired morphism. □